

Submission & Reviewer Matching

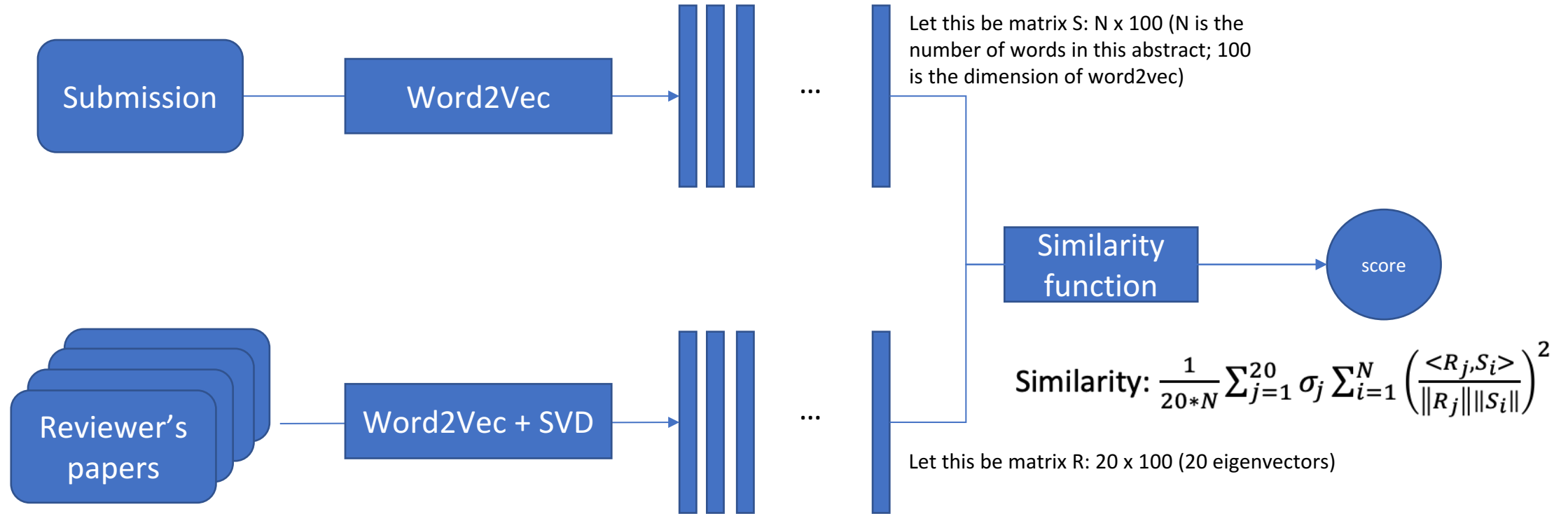
Weekly update from Qiang

08/06/2019

Update

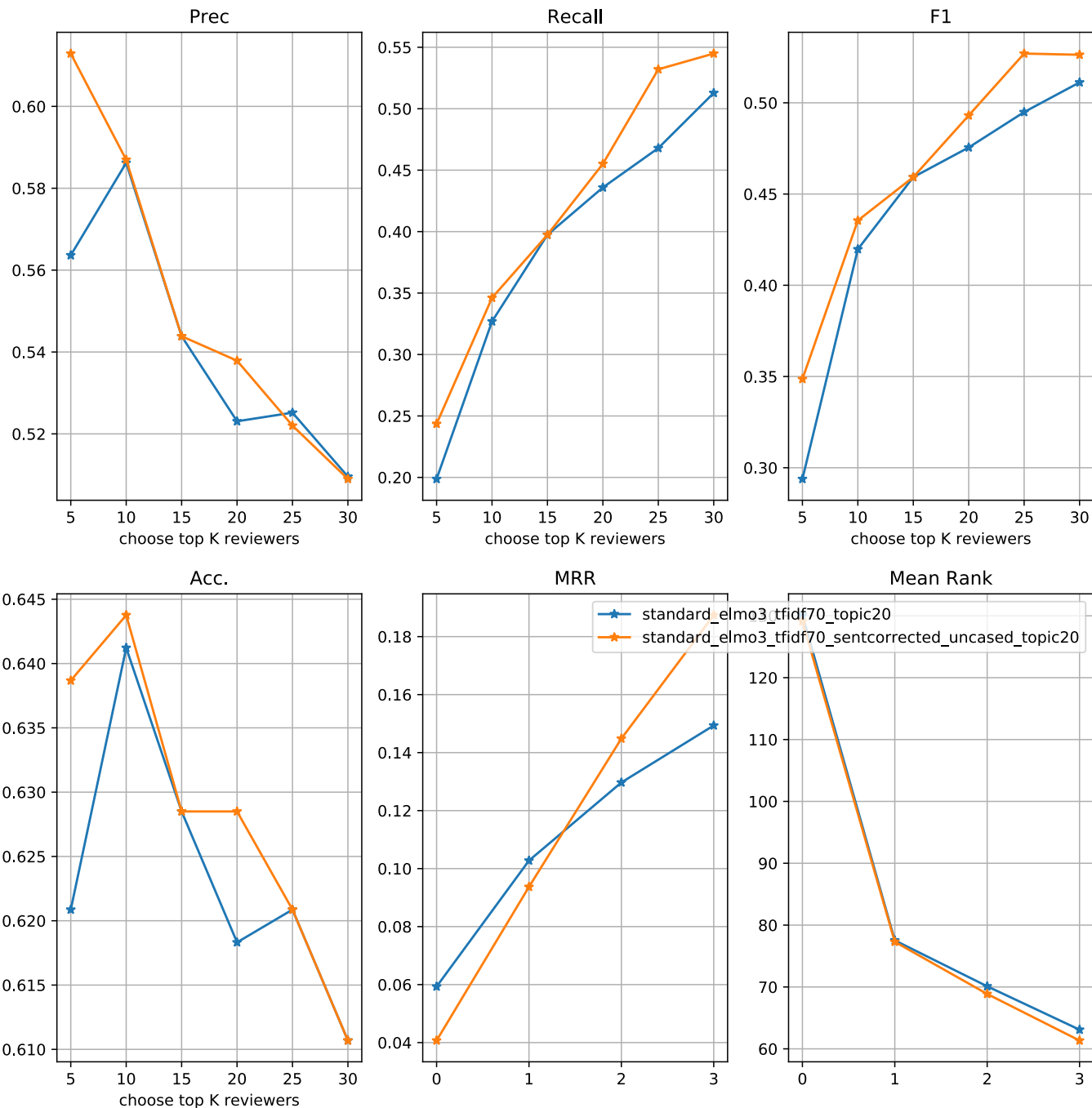
1. Initial try of TFIDF
2. Evaluation metric
3. Word embeddings
4. Topic weights
5. Re-organized the code
6. Summary of previous results
7. Initial try of ELMo
8. Retry ELMo
9. Proposal of next steps

Current system

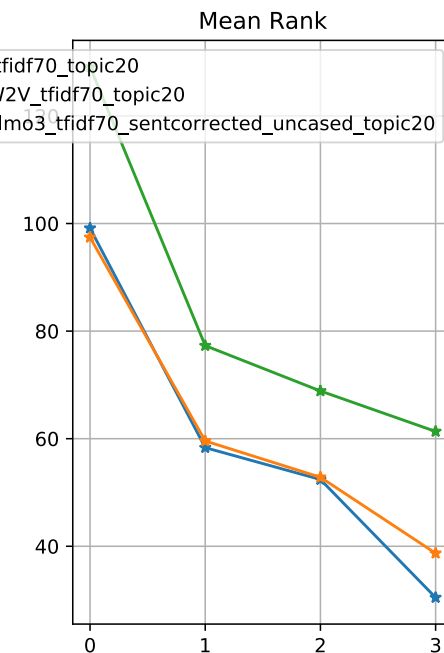
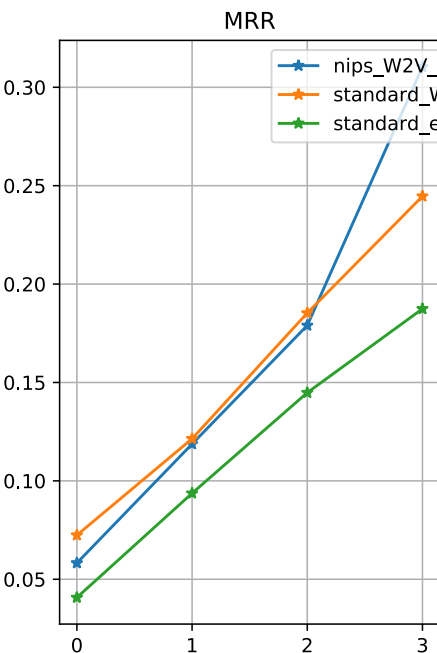
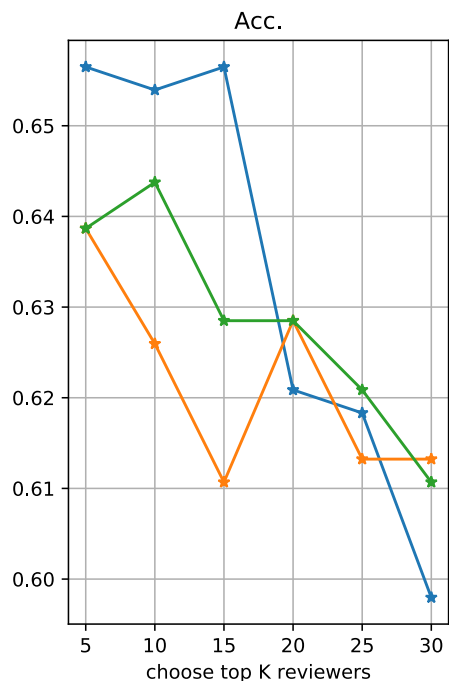
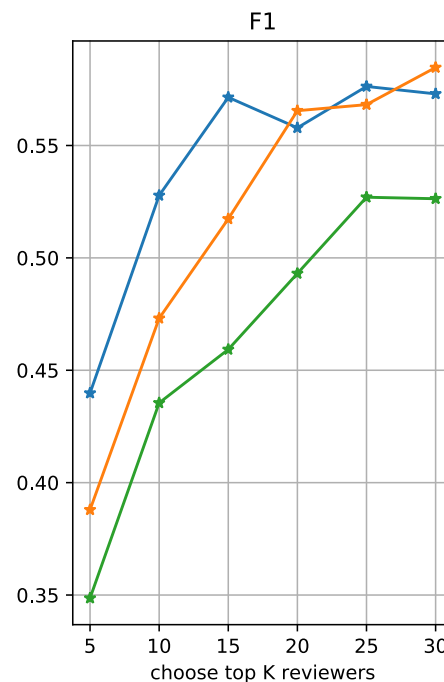
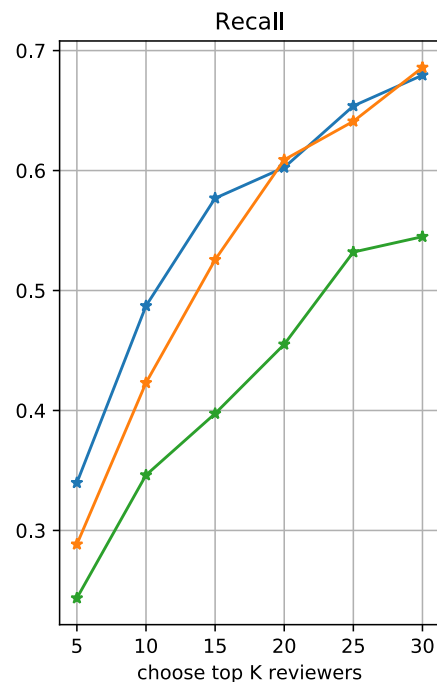
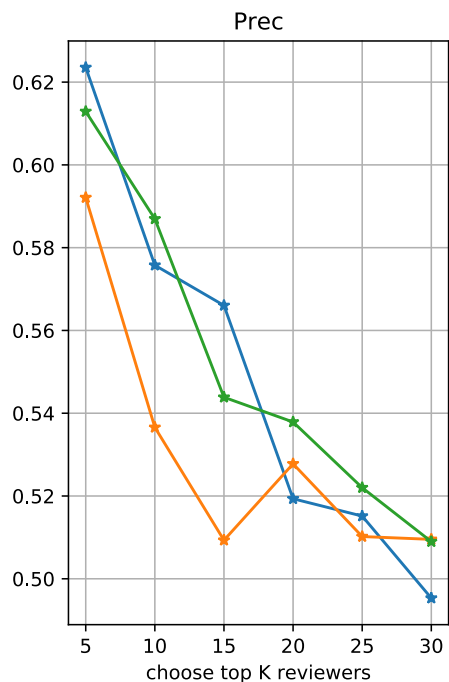


Why ELMo failed in the beginning

- “a key issue in supervised protein classification is the representation of input sequences of amino acids recent work using string kernels for protein data has achieved state-of-the-art classification performance however, such representations are based only on labeled data examples with known 3d structures, organized into structural classes while in practice, unlabeled data is far more plentiful in this work, we develop simple and scalable cluster kernel techniques for incorporating unlabeled data into the representation of protein sequences we show that our methods greatly improve the classification performance of string kernels and outperform standard approaches for using unlabeled data, such as adding close homologs of the positive examples to the training data we achieve equal or superior performance to previously presented cluster kernel methods while achieving far greater computational efficiency”



Orange line is consistently better than the blue line. This means after fixing the sentence splitting issue, ELMo achieved a better performance.



Green line is the best ELMo could achieve now (after fixing the sentence splitting issue), but it's still lower than the blue line.

Even when comparing to the orange line, the green line shows significant loss in recall and F1 and some improvement in precision and accuracy.

Summary of the conclusions we have so far

- Contextualized embeddings are not necessarily improving
- Keywords extraction is very helpful
- In-domain embeddings are better than off-the-shelf embeddings
- Evaluation metrics for ranking problems are suitable here.
 - $F1@K$, $Acc@K$, MRR
 - $NDCG$, Spearman's ρ , mean average precision
- Computational time: hundreds of reviewers and hundreds of papers should be solved within minutes on a single desktop (plus, a lot of computations can be done and saved before the assignment starts)